GlyGen AI-Assisted Biocuration

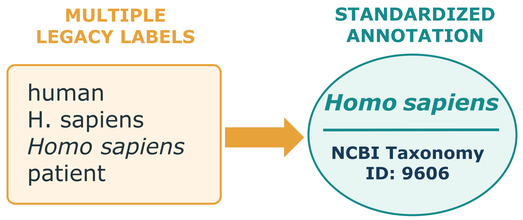
Species, Tissue, and Disease Annotation

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INTRODUCTION

Legacy biological datasets often contain unstructured free-text labels, including abbreviations, misspellings, informal terms, and outdated names. Because biological definitions and nomenclature change over time, these labels do not always map directly to current reference identifiers. This complicates integration with modern databases and risks losing scientific knowledge captured in legacy records.



Normalization links name variants to one reference record.

HYPOTHESIS

Can an LLM-assisted workflow normalize legacy biological labels and improve mapping accuracy to current reference identifiers through iterative prompt refinement?

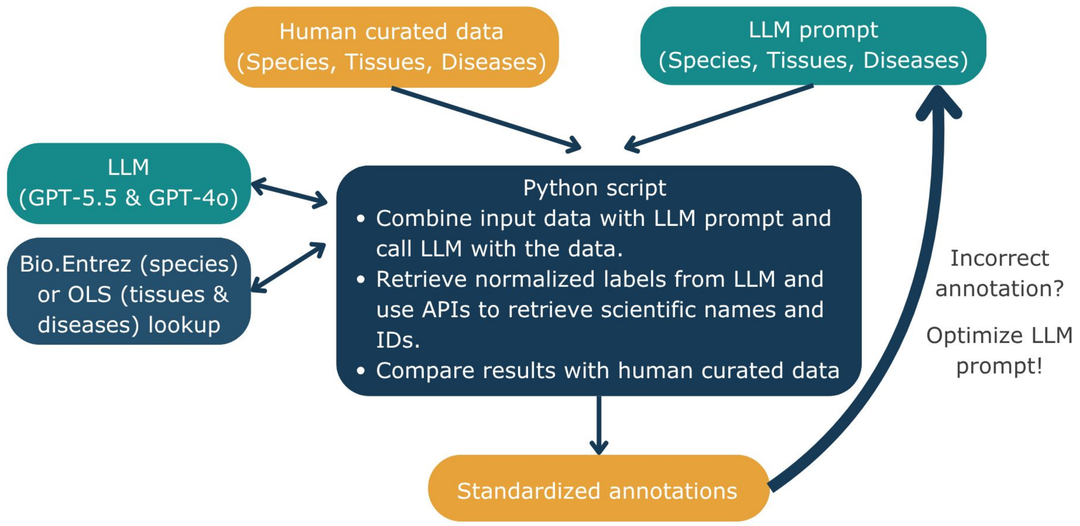
OBJECTIVES

1. **Normalize**
   Convert variable legacy labels into standardized biological names using an LLM.
2. **Map**
   Link normalized names to identifiers from NCBI Taxonomy, UBERON, or Disease Ontology.
3. **Evaluate**
   Compare automated mappings with curator-verified references and measure how prompt changes affect accuracy.

WHY IT MATTERS

- Faster conversion of legacy free-text records.
- Preserves scientific meaning as biological nomenclature changes.
- Consistent links to reference databases and ontologies.

PROJECT WORKFLOW

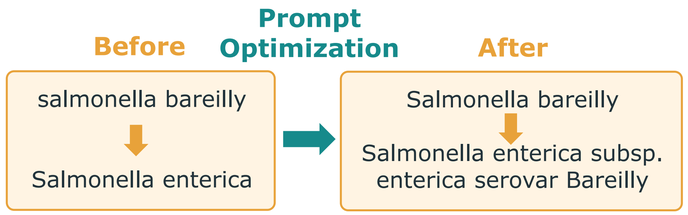


PROMPT ENGINEERING



KEY LESSONS LEARNED

- Give the LLM a role.
- Keep prompt concise.
- Use direct language.
- Be consistent.
- Ordered procedure section.
- Force "No Match Found".
- Optimize for no hallucinations.
- Add correct/failure examples.



DISCUSSION

- Difficult species names involving synonyms, outdated nomenclature, and taxonomic ambiguity were more challenging than direct matches. These names were previously mapped manually and therefore located in Figure 1.
- Model choice affected output quality, highlighting the importance of evaluating model selection and prompt design.
- To minimize token costs, GPT-4o was utilized during prompt engineering, whereas GPT-5.5 was used to generate the initial and final results.
- Hallucinations remain a concern, as LLMs may return plausible but incorrect scientific names or taxonomic assignments.
- Conservative matching prioritizes precision over recall: returning "No Match Found" avoids unsupported mappings but may leave valid matches unresolved.
- Curator review remains essential to verify biological meaning, resolve ambiguous cases, and confirm identifier accuracy.

SPECIES RESULTS

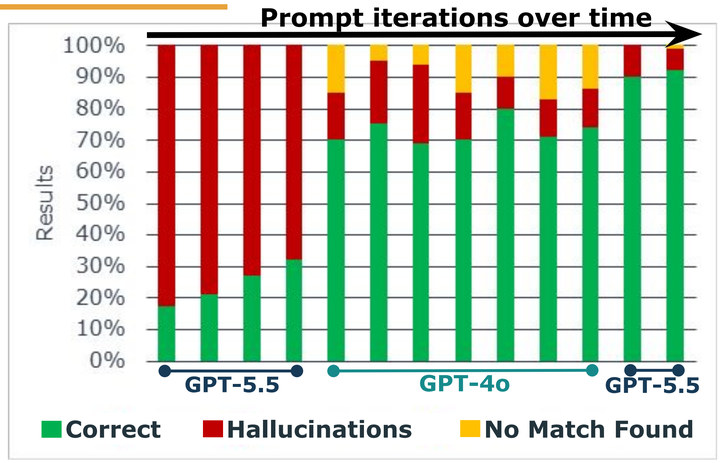


Figure 1. Results from dataset with species that needed to be mapped manually

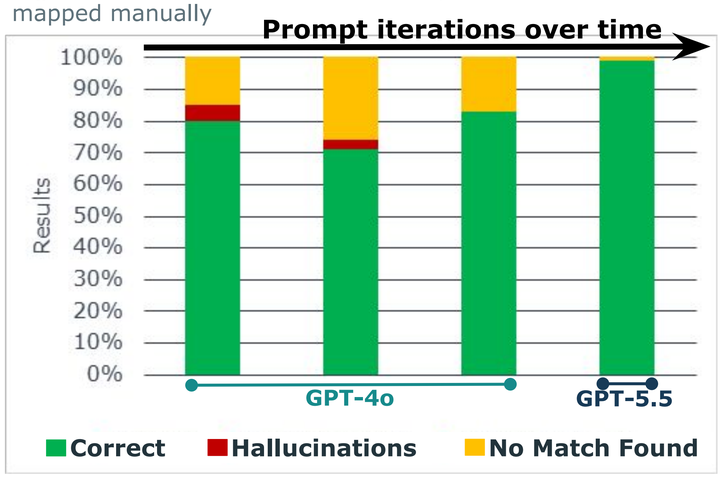


Figure 2. Results from dataset with species that did not need to be mapped manually

RESULTS EXPLAINED

- Prompt refinement improved identification
- Type of GPT model used effected result quality
- GPT-5.5 used initially
- Switched to GPT-4o during prompt development and iterative optimization for cost saving
- Final evaluation using GPT-5.5

FUTURE WORK

- Extend strategy and workflow for tissue and disease labels using own prompt optimizations.
- Continue to quantify the effects of prompt refinement on mapping accuracy.
- Test optimized prompts with different LLMs including smaller and free models.
- Use optimized prompts for mapping user provided labels to dictionary terms in data submission platforms such as GlyTableMaker.